

GIDEON STEIN

Machine Learning Researcher

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Gideon-Stein

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EDUCATION

PhD: Causal Discovery in Real-world Applications

FSU Jena 📅 May 2021 – Present 📍 Jena, Germany

- PhD project in cooperation with the CVG Jena and iDiv
- Focus on the construction of large-scale benchmarks for causal discovery with downstream applications in biodiversity research

M.Sc. Machine Learning & Data Analysis

ITMO University 📅 2018 – 2020 📍 St. Petersburg, Russia

- Independent research publications on Reinforcement Learning
- Awarded "Best Thesis" by ITMO University

B.A. Philosophy & Economics

University of Bayreuth 📅 2014 – 2018 📍 Bayreuth, Germany

- Focus on Game Theory and Decision Theory

SELECTED PUBLICATIONS

G. Stein, et al. "TCD-Arena: Assessing robustness of time series causal discovery against assumption violations". **ICLR 2026**.

G. Stein, et al. "CausalRivers – Scaling up benchmarking of causal discovery for real-world time-series". **ICLR Spotlight 2025**.

Yuan Yuan Huang* & G. Stein*, et al. "Enhanced stability of grassland soil temperature by plant diversity". **Nature Geoscience 2024**.

G. Stein, et al. "Embracing the black box: Heading towards foundation models for causal discovery from time series". **AAAI WS 2024**.

INDUSTRY AND RESEARCH

Research Associate: KI4KI - Remote Sensing for Large Infrastructure monitoring

FSU Jena 📅 May 2022 – Present 📍 Jena, Germany

- Research on satellite-based dam deformation forecasting

Freelance ML Consultant & Software Engineering

steinML / EasyMedBox / bliro.io / Desko 📅 Sep 2020 – Present

📍 Remote / Jena, Germany

- Machine learning consulting
- Feasibility studies on VLMs and ASR systems.
- Early-stage product development for tech startups (EasyMedBox, Bliro) parallel to doctoral research.

ABOUT ME

I am a Machine Learning Researcher and PhD candidate specializing in **Causal Discovery, Time Series Processing, and Deep Learning**. My vision is to build automated systems that extract causal structure from observational data, fundamentally rethinking how we uncover complex relationships in both the sciences and the private sector. Currently I am laying the groundwork for this by constructing large-scale benchmarks and working on foundation models.

SKILLS

Research Focus

Causal Discovery

Time-Series Processing

Deep Learning

Benchmarking

Reinforcement Learning

Frameworks and Tools

Python

PyTorch

PyTorch Lightning

Slurm

Docker

Hydra

Git

LaTeX

Open-CV

Scikit

Pandas

NumPy

LANGUAGES

German

Native

English

Full Professional (C2)

REFEREES

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